

Mobile Application Development with Flutter

Lecture 54. Flutter APIs & REST

What is an API?

- API stands for “Application Programming Interface”
- An API allows our Flutter app to communicate with a server/database
- Example:
 - Our app asks: “Give me latest news”
 - Server replies with data
- Think of API as a waiter between:
 - Our app (client)
 - Server/database

Example in Mobile Apps

- WhatsApp
 - When we open WhatsApp:
 - App requests messages from API
 - Server sends messages back
- YouTube
 - When videos load YouTube app calls APIs
 - Server returns:
 - Titles
 - Thumbnails
 - Views
 - Comments

Why APIs are Important?

- Without APIs:
 - Apps cannot access internet data
 - No login system
 - No live content
 - No cloud storage
- Modern apps are heavily API-based.

Client-Server Architecture

- Client
 - The client is
 - Flutter app, Website, Desktop app
 - Its job:
 - Request data
 - Display data to user
- Servers
 - The server:
 - Stores data, Processes requests, Sends responses
 - Examples:
 - Facebook servers, Google servers, Netflix servers

Communication Flow

- Flutter App → API Request → Server
- Flutter App ← API Response ← Server

`</> JSON`

```
[
  {
    "username": "Ali",
    "caption": "Hello"
  }
]
```

API vs REST API

- An **API** is a set of rules that allows two software programs to communicate with each other
- For example accessing Google Maps or Payment Processor without understanding how internal code works
- REST stands for Representational State Transfer
- A **REST API** is a specific type of API that follows a set of design principles known as REST

REST API Constraints

- **Statelessness**
 - The server doesn't store any information about past requests
 - Every request must contain all the information needed to process it
- **Uniform Interface**
 - It uses standard **HTTP** methods to perform actions (GET, POST, PUT/PATCH, DELETE)
- **JSON Format**
 - While it can support XML or HTML, most modern REST APIs use **JSON**
 - JSON is lightweight and easy for both humans and machines to read
- **Client-Server Separation**
 - The user interface (client) and the data storage (server) are separate, allowing them to be updated or scaled independently

Key Differences at a Glance

Feature	General API	REST API
Scope	Any interface for software communication	A specific type of API following REST rules
Protocol	Can use any (HTTP, gRPC, SOAP, etc.)	Primarily uses HTTP
Data Format	Any (XML, Binary, JSON, etc.)	Usually JSON
State	Can be stateful or stateless	Must be stateless

REST API Structure

- Example:
 - <https://api.example.com/users>
- Explanation:

Part	Meaning
https	Protocol
api.example.com	Server
/users	Resource

What is HTTP?

- HTTP stands for:
 - HyperText Transfer Protocol
- It is the communication system of the web
- When Flutter talks to server:
 - it uses HTTP requests

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HTTP Methods

- Methods define WHAT action we want
- GET Method
 - Its purpose is to fetch data
 - GET is the most common method
 - Examples:
 - Get users, Get products, Get posts
- POST Method
 - Its purpose is to send/create new data
 - Examples:
 - Create account, Add comment, Upload post

HTTP Methods

- PUT Method
 - Its purpose is to update existing data
 - Examples:
 - Update profile, Edit product
- DELETE Method
 - Its purpose is to remove data
 - Examples:
 - Delete messages
 - Remove products

Easy Analogy (CRUD Operations)

Action	HTTP Method
Read	GET
Create	POST
Update	PUT
Delete	DELETE

What is JSON?

- JSON is the MOST IMPORTANT topic in APIs
- JSON stands for “JavaScript Object Notation”
- It is the standard format for sending data
- Why JSON is Used
 - Lightweight
 - Easy to read
 - Fast
 - Supported everywhere

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JSON Format

- JSON Object, JSON Array, Nested JSON

```
{  
  "name": "Ali",  
  "age": 22  
}
```

```
[  
  {  
    "name": "Ali"  
  },  
  {  
    "name": "Ahmed"  
  }  
]
```

```
{  
  "user": {  
    "name": "Ali",  
    "city": "Lahore"  
  }  
}
```

Status Codes

- When server responds it sends status code
- Most important status codes

Code	Meaning
200	Success
201	Created
400	Bad request
401	Unauthorized
404	Not found
500	Server error

API Testing Tools

- Before Flutter coding, developers often test APIs using:
 - Postman
 - Browser
 - Thunder Client
- Example Public API
 - Open this in browser:
 - <https://jsonplaceholder.typicode.com/posts>
 - We'll see JSON data
 - This is free practice API

Synchronous vs Asynchronous

- Synchronous
 - Tasks happen one by one
- Asynchronous
 - Flutter can wait for internet response WITHOUT freezing app
 - This is why APIs use:
 - Future, async, await
- Why API calls take time
 - Because data travels through:
 - Internet, routers, servers, databases
 - This delay is called latency